

LI-FI BASED SMART TRAFFIC NETWORK

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ABSTRACT

Cities are struggling to cope with problems caused by the increased demand of roadway. In a future where autonomous cars replace our current human based traffic system, the smart traffic system is designed to allow communication between automobiles and their traffic environment. This is usually called V2X technology. Using Li-Fi, a visible light communications system, autonomous cars will be able to communicate with each other and with the traffic network to eradicate any human error that might occur while driving today. That said, this Li-Fi based system is complimentary to existing autonomous systems that cars use such as radar, LIDAR or cameras. Using light pulses at a specific frequency, an FPGA will be used to send a unique signal for every traffic light, sign and car.

Index Terms—Li-Fi, FPGA, LIDAR, UART,

I. INTRODUCTION

THE Smart Traffic System is designed to allow stop signs, traffic signals, and even roadways to communicate with automobiles to provide the fastest, smoothest and safest ride for passengers. Modern autonomous cars operate by using radar, Global Positioning Systems (GPS), computer vision, odometry, or a combination of the aforementioned technologies. A common technology that utilizes radio waves is Vehicle-to-Vehicle (V2V) and Vehicle-to-infrastructure (V2I), collectively known as V2X, which allows automobiles to communicate with other automobiles and their traffic environment. [1] V2V and V2I uses the same frequency as Wi-Fi. Cable companies and other technology industries are fighting to take away part of the radio spectrum that V2V uses. Any new technology that uses radio waves will have to compete with existing technologies that currently have control of the radio spectrum. The Smart Traffic System will utilize Li-Fi to avoid using the already crowded radio air waves. Our system will also eliminate a problem faced by the Tesla Autopilot System, a currently popular autonomous car technology. [2] This system is currently designed to be used on limited-access highways, and not in urban streets as it is unable to improvise in construction areas and locations with no lane markings. Our system would require a light transmitter be placed on each traffic sign to send a signal to cars that alert the system of the position of each sign so the car can determine the action required by each sign. As traffic signals already emit light, they would not require an additional transmitter. They would only need to be reprogrammed in order to change the flash rate of the existing lights.

II. BUILDING IT UP

This project was simulated using Intel's Altera Cyclone V FPGA. The Field Programmable Gate Array (FPGA) was chosen for its versatility, usability, and flexibility which is ideal for prototyping. The Cyclone V has many improvements compared to its earlier generations such as its low power usage. The Altera Cyclone V features an ARM based 1.6GHz base frequency, Intel dual-core Arm Cortex A9 MP Core processor. Each processor provides 32 KB of L1 instruction cache along with 32 KB of data cache to make up a 64-Bit instruction set, 512 KB of L2 cache, 64 KB of scratch RAM, and an 8 channel direct memory access controller. The powerful Intel processor also boasts 2.5 GT/s DMI and a 3.5 Watt TDP. [3] These processors contain a rich set of peripherals, and a multiport memory controller which is shared with logic to create flexibility and cost savings. FPGA resources are freed due to these hardened peripherals, since the need to implement functions in the program are no longer required. Intel's Quartus Prime II software was used as the simulation medium for the project. Three separate sets of code were needed to fully simulate a basic version of the project. Using the Universal Asynchronous Receiver-Transmitter (UART) capability of the Cyclone, a transmitter code and a receiver code are used for the brains behind the system, with a test bench code used to monitor how the system acts. [4] The codes were written in VHDL since a hardware description language is needed to program the Cyclone. The transmitter code and receiver code are very similar in context so only a portion of the transmitter code is shown below in Figure 1.

```
1 library IEEE;
2 use IEEE.Std_logic_1164.ALL;
3 use IEEE.Numeric_std.all;
4
5 Entity UART_RX is
6   generic (
7     g_CLKS_PER_BIT : integer := 115
8   );
9   port (
10    i_Clk : in std_logic;
11    i_RX_Serial : in std_logic;
12    o_RX_DV : out std_logic;
13    o_RX_Byte : out std_logic_vector(7 downto 0);
14  );
15 end UART_RX;
16
17 architecture rtl of UART_RX is
18   type t_SM_Main is (s_Idle, s_RX_Start_Bit, s_RX_Data_Bits,
19                     s_RX_Stop_Bit, s_Cleanup);
20   signal r_SM_Main : t_SM_Main := s_Idle;
21   signal r_RX_Data_R : std_logic := '0';
22   signal r_RX_Data : std_logic := '0';
23   signal r_Clk_Count : integer range 0 to g_CLKS_PER_BIT-1 := 0;
24   signal r_Bit_Index : integer range 0 to 7 := 0;
25   signal r_RX_Byte : std_logic_vector(7 downto 0) := (others => '0');
26   signal r_RX_DV : std_logic := '0';
27
28 begin
29   p_SAMPLE : process (i_Clk)
30   begin
31     if rising_edge(i_Clk) then
32       r_RX_Data_R <= i_RX_Serial;
33       r_RX_Data <= r_RX_Data_R;
34       end if;
35   end process p_SAMPLE;
36
37   p_UART_RX : process (i_Clk)
38   begin
39     if rising_edge(i_Clk) then
40       case r_SM_Main is
41         -- s_Idle
42         when s_Idle =>
43           r_Clk_Count <= 0;
44           r_Bit_Index <= 0;
45           r_RX_Byte <= (others => '0');
46           r_RX_DV <= '0';
47           r_SM_Main <= s_RX_Start_Bit;
48         -- s_RX_Start_Bit
49         when s_RX_Start_Bit =>
50           r_Clk_Count <= 0;
51           r_Bit_Index <= 0;
52           r_RX_Byte <= (others => '0');
53           r_RX_DV <= '0';
54           r_SM_Main <= s_RX_Data_Bits;
55         -- s_RX_Data_Bits
56         when s_RX_Data_Bits =>
57           r_Clk_Count <= 0;
58           r_Bit_Index <= 0;
59           r_RX_Byte <= (others => '0');
60           r_RX_DV <= '0';
61           r_SM_Main <= s_RX_Stop_Bit;
62         -- s_RX_Stop_Bit
63         when s_RX_Stop_Bit =>
64           r_Clk_Count <= 0;
65           r_Bit_Index <= 0;
66           r_RX_Byte <= (others => '0');
67           r_RX_DV <= '0';
68           r_SM_Main <= s_Cleanup;
69         -- s_Cleanup
70         when s_Cleanup =>
71           r_Clk_Count <= 0;
72           r_Bit_Index <= 0;
73           r_RX_Byte <= (others => '0');
74           r_RX_DV <= '0';
75           r_SM_Main <= s_Idle;
76       end case;
77     end if;
78   end process p_UART_RX;
79 end rtl;
```

Fig. 1. UART Transmitter Code in VHDL: The above code

is a set up for the transmitter; it will dictate what start bit to initiate sending data to the receiver.

III. BREAKING IT DOWN

Once the start bit signal is received, the FPGA is open to the data signal coming from the transmitter side. The stop bit signifies the end of the data transmission. This is then compared with the cache memory to understand what the signal is conveying, whether it is to stop the car, slow down, speed up or any number of the countless cases that can occur on the road. The simulation was run under 8-bit data signals but real life application of this concept would require at least 128 or 256 bits to accommodate all the different traffic signs, lights, and vehicles on the road and hexadecimal encryption and additional extra layers of encryption for security.

```

44 signal r_CLOCK      : std_logic      := '0';
45 signal r_TX_DV      : std_logic      := '0';
46 signal r_TX_BYTE     : std_logic_vector(7 downto 0) := (others => '0');
47 signal w_TX_SERIAL   : std_logic;
48 signal w_TX_DONE     : std_logic;
49 signal w_RX_DV       : std_logic;
50 signal w_RX_BYTE     : std_logic_vector(7 downto 0);
51 signal r_RX_SERIAL   : std_logic := '1';
52
53
54 -- Low-level byte-write
55 procedure UART_WRITE_BYTE (
56   i_data_in : in std_logic_vector(7 downto 0);
57   signal o_serial : out std_logic) is
58 begin
59
60   -- Send Start Bit
61   o_serial <= '0';
62   wait for c_BIT_PERIOD;
63
64   -- Send Data Byte
65   for ii in 0 to 7 loop
66     o_serial <= i_data_in(ii);
67     wait for c_BIT_PERIOD;
68   end loop; -- ii
69
70   -- Send Stop Bit
71   o_serial <= '1';
72   wait for c_BIT_PERIOD;
73 end UART_WRITE_BYTE;
74
75
76 begin
77
78   -- Instantiate UART transmitter
79   UART_TX_INST : uart_tx
80   generic map (
81     c_CLKS_PER_BIT => c_CLKS_PER_BIT
82   )
83   port map (
84     i_clk      => r_CLOCK,
85     i_tx_dv    => r_TX_DV,
86     i_tx_byte  => r_TX_BYTE,
87     o_tx_active => open,
88     o_tx_serial => w_TX_SERIAL,
89     o_tx_done  => w_TX_DONE
90   );

```

Fig. 2. UART Test bench code in VHDL: It is essential to note that the following code initiates with a start bit and when the receiver detects the start bit it will begin to process the rest of the incoming data from the transmitter.

The code in Figure 2 tests the combination of the transmitter and receiver codes. The code for the test bench is a simple code that sends the 10-bit signal (one start bit, one stop bit and the 8-bit data) and check whether the data transmitted is the same as data received. Since the system is asynchronous, the receiver would detect the start bit to initiate processing the data coming to it.

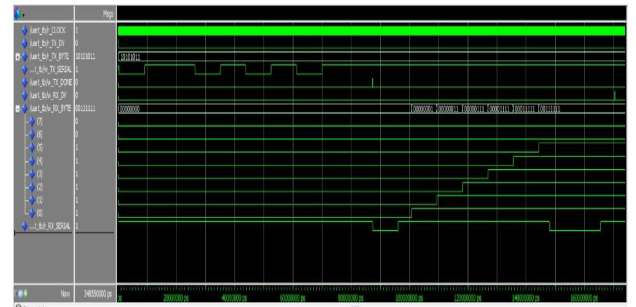


Fig. 3. UART simulation results: the following is the result of the simulation starting with the transmitter sending the test bit and the receiver starting to process the data.

It took about 175 us for the simulation to send the 8 bit data successfully. With 256 bits, the whole process would take about 2.5 ms for the data to be transmitted. That means for a car going at 80mph, it would take about 3.5 inches to send a signal from the traffic sign to the car and another 3.5 inches to reply and acknowledge the signal, possibly indicating to slow down or switch lanes in case of a construction zone obstruction known by the system. At 670,600,000 miles per hour, nothing travels faster than the speed of light, but it does take a small amount of time to process the signal. In the real application, the transmitting and receiving time will be a lot faster due to the increased clock rate and processing speed. There has been a lot of talk regarding the safety and security of the system.

IV. POSSIBLE HURDLES

As mentioned earlier, the Li-Fi network will work simultaneously with other functions included in an autonomous car, and this network will be monitored by the local traffic department[5]. The system in the vehicle will encounter many obstacles while it's operational. Other traffic signs in the area might interfere with the signal from the significant signs in a specific location. Using GPS systems which is already incorporated in almost all modern cars, the location of the vehicle and the signs are to be recorded in the network mainframe so that the vehicle would expect a signal from the signs in its way and ignore all the other signals from irrelevant traffic signs and traffic lights. Alien obstacles however, might block the signal between two communicating objects, in this case the current technologies in autonomous vehicles would be relied on, such as Lidar, radar, and cameras scanning the surroundings of the vehicle to navigate safely.

A. SURFACE APPLICATIONS

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B. EFFECT OF WEATHER CONDITIONS

Some might argue that the system is not as fast and reliable as the current standard in autonomous cars. Currently, autonomous vehicles process a huge amount of information the Lidar system since it scans the surrounding area point by point [6]. Li-Fi on the other hand would just receive the signal of what's ahead and act accordingly using the cameras. With processors becoming faster and more efficient, they will be able to process the information received by Lidar in a fraction of the time of past technologies. With Li-Fi working with the cameras and Lidar system, the reaction time would not be affected in comparison to current times and would probably be much faster as well.

Some concerns of a fully automated vehicle is the effect the weather has on the systems. New radar technologies have found a way around this issue. Some monitoring systems use low-frequency radar which is usually less than one gigahertz, which works fine when the weather is clear. Light has trouble passing through fog, that's where an alternative is needed. Modern systems use higher frequency radar in the order of gigahertz to detect obstacles [7]. Since fog scatters light, low frequencies will have a small phase shift which would corrupt the signal. Higher frequency would result in an even greater phase shift which would eventually cancel out the scattered signals and receive only the original signal at the detector [8]. It is important to note that the computational algorithm is also crucial in the operation of the system. Some new techniques using Lidar are still in their primitive phase but can detect objects in thick fog. Even though the light will be reflected from both the object and the fog, an advanced algorithm uses statistics to separate between the two and calculate the object's distance [9].

V. CONCLUSIONS AND THE FUTURE OF TRANSPORTATION

The concept of the autonomous car has repeatedly been introduced to society, however many drivers feel that a program error or virtual hijacking can happen which can result in accidents. Along with the impression that society has towards autonomous vehicles, the system cannot be implemented within a short amount of time. It will take many years to implement the devices due to economic factors and politics. Also, the system on vehicle cannot simply rely on current autonomous technology alone. The addition of Li-fi sensors on the road and on signs will greatly enhance the capability of the autonomous vehicle, and evolve the driver assistance technology into a driverless era. There will be more obstacles in the future, but with time the smart traffic system can become a reality. The full paper will include detailed

analysis of the system and a more in-depth view of the system working with current technologies.

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